
1.2. Examples of Fields that use Digital Processing:

A digital image is a representation of a two-dimensional image as a finite set of digital values, called picture elements or pixels.

Following are the field where Digital image processing are used

- I. Gamma Rays
- II. X-Rays
- III. Ultraviolet Rays
- IV. Visible and Infrared Rays
- V. Microwave Rays
- VI. Radio Band

I. Gamma rays :

- Gamma rays are used in Medicine and Astronomical Observations.
- It is also used in Bone Pathology.
- It is used to find the Tumors and Infections in our body.
- In Nuclear Medicine, Rays are passed into our body with a radioactive isotope that emits gamma rays as it decays.
- The images are produced and they are collected by Gamma Ray Detectors.

II. X-Rays:

- It is used widely used in
- Medical Diagnostics
- Astronomy
- In Medical, Rays are generated using X-Rays tube, which is a Vacuum Tubes with a Cathode and Anode.
- ANGIOGRAPHY, is used to obtain images about the Blood Vessels.
- Higher energy X-rays are used in Industrial processes.

III. Ultraviolet rays:

UV rays are widely used in

- Lithography
- Industrial Inspection
- Microscopy
- Lasers
- Biological Imaging
- Astronomical Observations
 - UV lights is us in Fluorescence microscopy.

- It is one of the fastest growing areas of Microscopy.
- Fluorescence is a method discovered in the middle nineteenth century.
- The ULTRAVIOLET Rays are not visible, when photon of ULTRAVIOLET radiations collides with an electron in an atom of fluorescent produces electrons with lots of energy.

IV. Infrared and visible rays/bands:

It plays an important roles such as

1. Microscopy 2. Astronomy 3. Remote Sensing 4. Industry 5. Law Enforcement

- ✓ Microscope allows us to see any tiny images with a well detailed form.
- ✓ Even at times objects viewed through a Microscope will not provide any clear ideas about the objects.
- ✓ To provide a competition to it and view in a well detailed manner another major area is used in visual processing called Visual and Infrared rays.

V. Microwave Rays:

The dominant application of imaging in the Microwave band is Radar.

- The very important feature of imaging Radar is its ability to collect data over virtually any region Radar at any time in any Weather or light condition.
- Some Radar waves can penetrate through clouds and also penetrate through vegetation and icy regions.

How Radar waves are generated?

- ✓ Instead of a camera lens, a radar uses an antenna and Digital Computer Processing to record images.
- ✓ In a Radar image, one can see only a Microwave energy that reflected from the Radar Antenna.

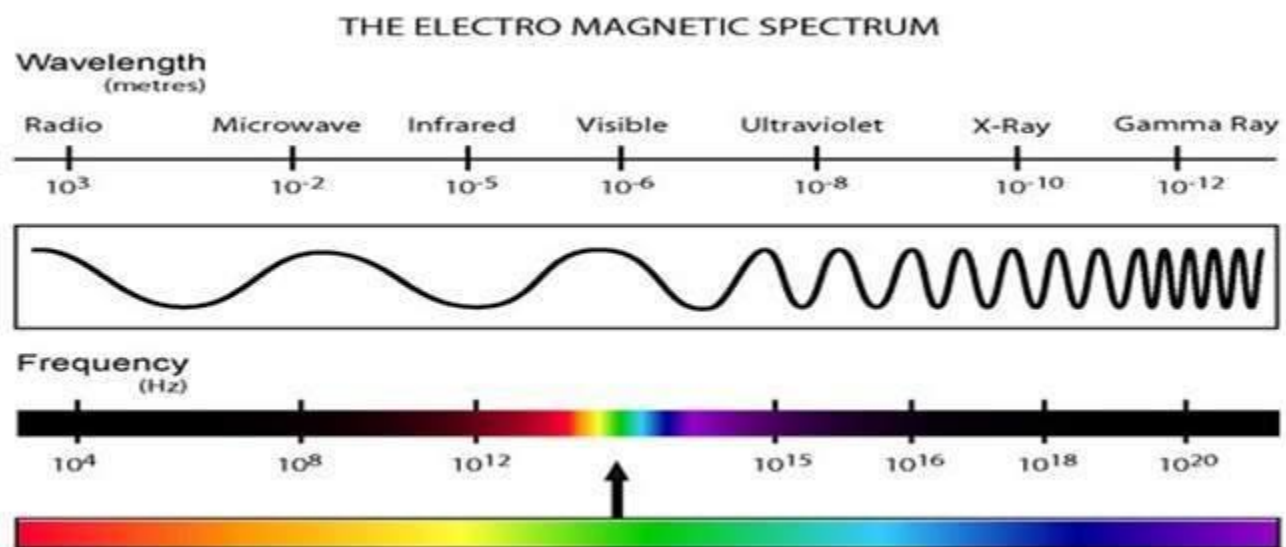
VI. Radio Rays/Bands:

Major uses of Radio bands:

- Astronomy :
- Medicine :

- In Medicine, Radio Waves are used in Magnetic Resonance Imaging (MRI).
- This technique places the patients in a powerful magnet and passes Radio waves through their body in very short pulses (in seconds).
- Each pulse causes a responding pulse of radio waves to be emitted by the patient's tissues.
- The location is then identified by the computer, which produces a 2-Dimensional picture.
- These MRI's can be used to take pictures in any plane.

The electromagnetic spectrum according to the energy of photon is shown below.



1.3 Application of Image Processing:

Some of the major fields in which digital image processing is widely used are mentioned below

- Image sharpening and restoration
- Medical field
- Remote sensing
- Transmission and encoding

- Machine/Robot vision
- Color processing
- Pattern recognition
- Video processing
- Microscopic Imaging
- Others

➤ **Image sharpening and restoration:**

Image sharpening and restoration refers here to process images that have been captured from the modern camera to make them a better image or to manipulate those images in way to achieve desired result. It refers to do what Photoshop usually does.

This includes Zooming, blurring, sharpening, gray scale to color conversion, detecting edges and vice versa, Image retrieval and Image recognition. The common examples are:

➤ **Medical field:**

The common applications of DIP in the field of medical is

- Gamma ray imaging
- PET scan
- X Ray Imaging
- Medical CT
- UV imaging

✓ **UV imaging:**

In the field of remote sensing, the area of the earth is scanned by a satellite or from a very high ground and then it is analyzed to obtain information about it. One particular application of digital

image processing in the field of remote sensing is to detect infrastructure damages caused by an earthquake.

As it takes longer time to grasp damage, even if serious damages are focused on. Since the area effected by the earthquake is sometimes so wide, that it not possible to examine it with human eye in order to estimate damages. Even if it is, then it is very hectic and time consuming procedure. So a solution to this is found in digital image processing. An image of the affected area is captured from the above ground and then it is analyzed to detect the various types of damage done by the earthquake.

The key steps include in the analysis are

- The extraction of edges
- Analysis and enhancement of various types of edges

➤ **Transmission and encoding:**

The very first image that has been transmitted over the wire was from London to New York via a submarine cable. The picture that was sent took three hours to reach from one place to another.

Now just imagine, that today we are able to see live video feed, or live CCTV footage from one continent to another with just a delay of seconds. It means that a lot of work has been done in this field too. This field does not only focus on transmission, but also on encoding. Many different formats have been developed for high or low bandwidth to encode photos and then stream it over the internet or etc.

➤ **Machine/Robot vision:**

Apart from the many challenges that a robot face today, one of the biggest challenge still is to increase the vision of the robot. Make robot able to see things, identify them, identify the hurdles etc. Much work has been contributed by this field and a complete other field of computer vision has been introduced to work on it.

➤ **Hurdle detection:**

Hurdle detection is one of the common task that has been done through image processing, by identifying different type of objects in the image and then calculating the distance between robot and hurdles.

➤ **Line follower robot:**

Most of the robots today work by following the line and thus are called line follower robots. This help a robot to move on its path and perform some tasks. This has also been achieved through image processing.

➤ **Color processing:**

Color processing includes processing of colored images and different color spaces that are used. For example, RGB color model, YCbCr, HSV. It also involves studying transmission, storage, and encoding of these color images.

➤ **Pattern recognition:**

Pattern recognition involves study from image processing and from various other fields that includes machine learning (a branch of artificial intelligence). In pattern recognition, image processing is used for identifying the objects in an image and then machine learning is used to train the system for the change in pattern. Pattern recognition is used in computer aided diagnosis, recognition of handwriting, recognition of images etc.

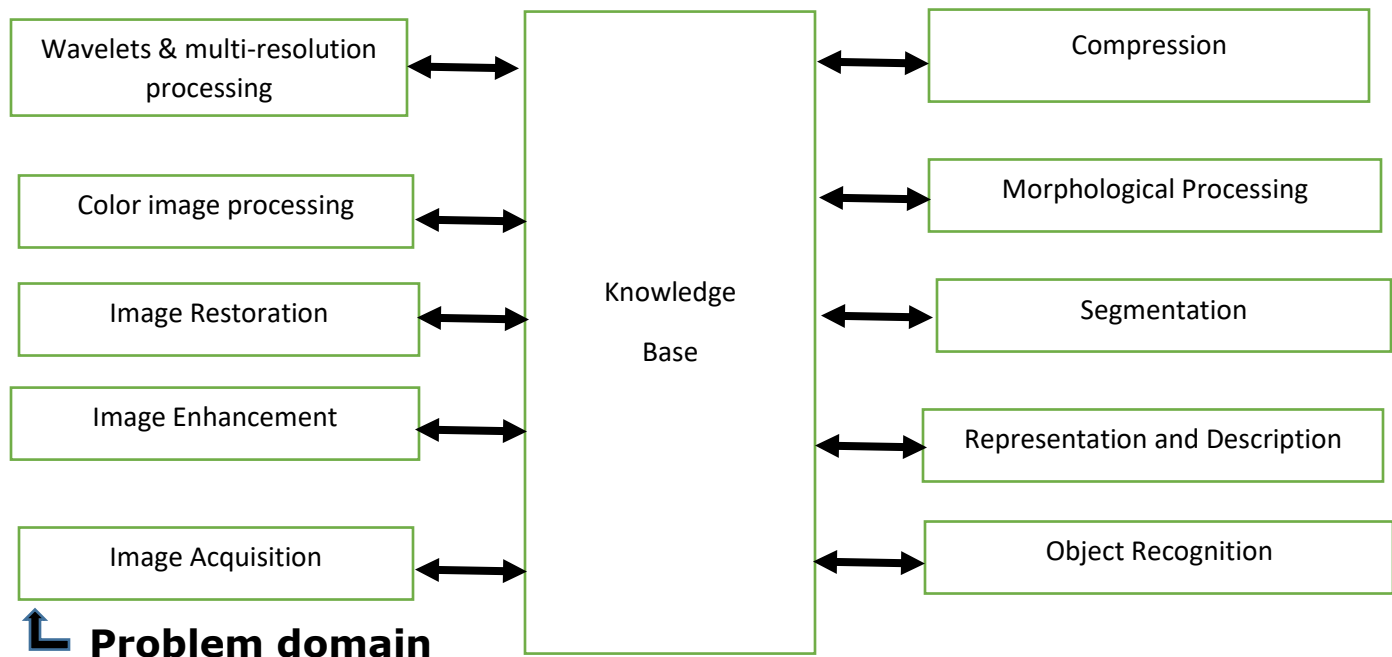
➤ **Video processing:**

A video is nothing but just the very fast movement of pictures. The quality of the video depends on the number of frames/pictures per minute and the quality of each frame being used. Video processing involves noise reduction, detail enhancement, motion detection, frame rate conversion, aspect ratio conversion, color space conversion etc.

➤ **Microscopic Imaging:**

- ✓ Microscope allows us to see any tiny images with a well detailed form.
- ✓ It enlarges the tiny images to provide with the full details.

1.4 Fundamental Steps in Image Processing Systems:



(Fig: Steps in Digital Image Processing)

I. Image Acquisition:

It is the first process in which preprocessing activities like scaling and rotating are done. It is the creation of a digitally encoded representation of the visual characteristics of an object, such as a physical scene or the interior structure of an object.

II. Image Enhancement:

In this stage, the original image is processed in such a manner that the resultant image will be more suitable with compared to original image.

III. Image Restoration:

It is the area that also deals with the improving of the quality or appearance of an image. Basically, image enhancement is subjective process but image restoration is objective process.

IV. Color image Processing:

With the help of Color, object identification and extraction from the scene is processed. Different types of color layer are separated from each other for further process.

V. Wavelets and Multi-resolution Processing:

Foundation for the representing image in various degree of resolution deals to the wavelets and multi resolution. Sometimes it is also used for image compression.

VI. Compression:

It deals with the technique for reducing the storage required to save an image to transmit. It deals with the reducing of the size of the image.

VII. Morphological Processing:

It is the combination of structure. So it deals with tools for extraction of an image component that area useful in representation and description of shape.

VIII. Segmentation:

It deals with the partition of an image into different parts and objectives.

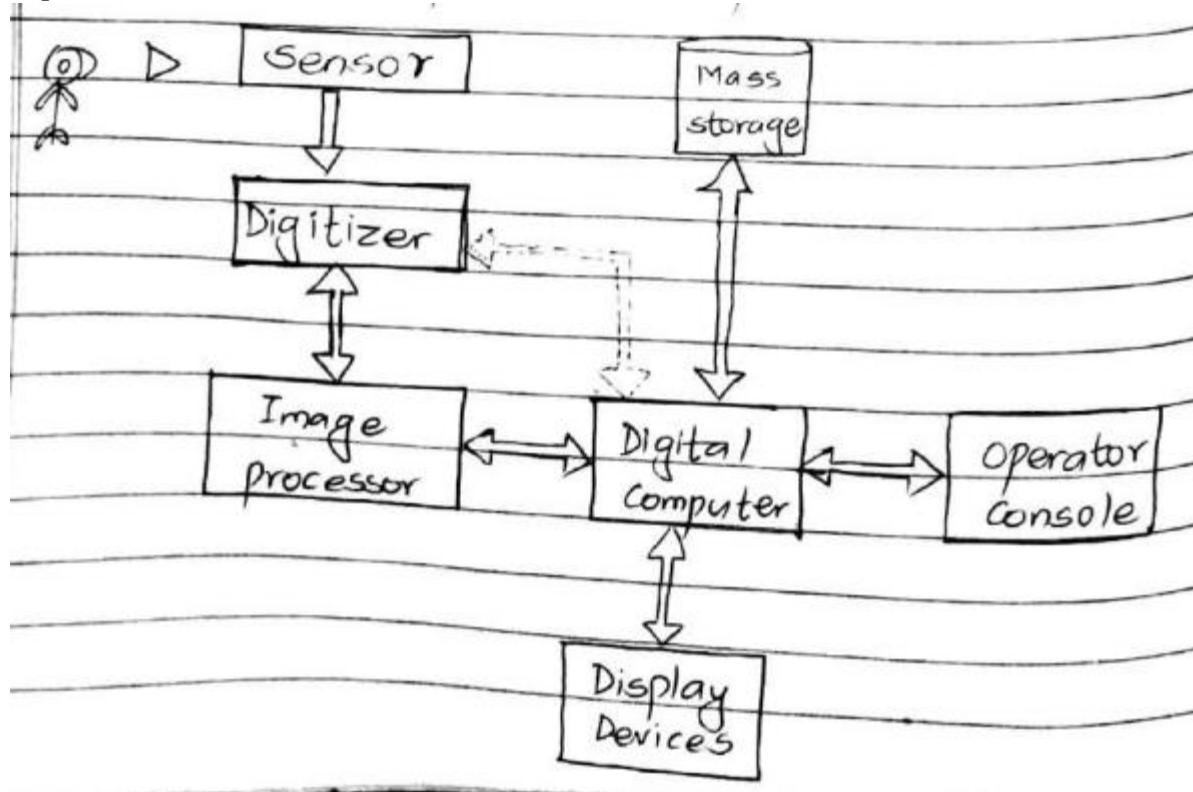
IX. Representation and Description:

After an image has been segmentation into sub regions then the resultant segmented pixels is representation and described in a suitable form for further analysis.

X. Object Recognition:

It processes that the level to an object based on its descriptors. It finally finds out what is the object was the image of.

1.5 Elements / Components of Digital Image Processing Systems:



(Fig: Elements of digital Image Processing)

i. Sensor:

It intercepts the energy propagating from the scene and transform it to produce an intensity image. Camera and scanner are the example of photo sensor.

ii. Digitizer:

The digitizer produces an image by the digitization technique. Normally, a digitizer converts an intensity image into numerical representation which is suitable for input into a digital computer.

iii. Image Processors:

It interfaces with digital computer to provide easy of programming for this purpose we can use microcomputer or mainframe computer.

Specially, it does functions like images acquisition, low level processing, scaling and rotating.

iv. Digital Computer:

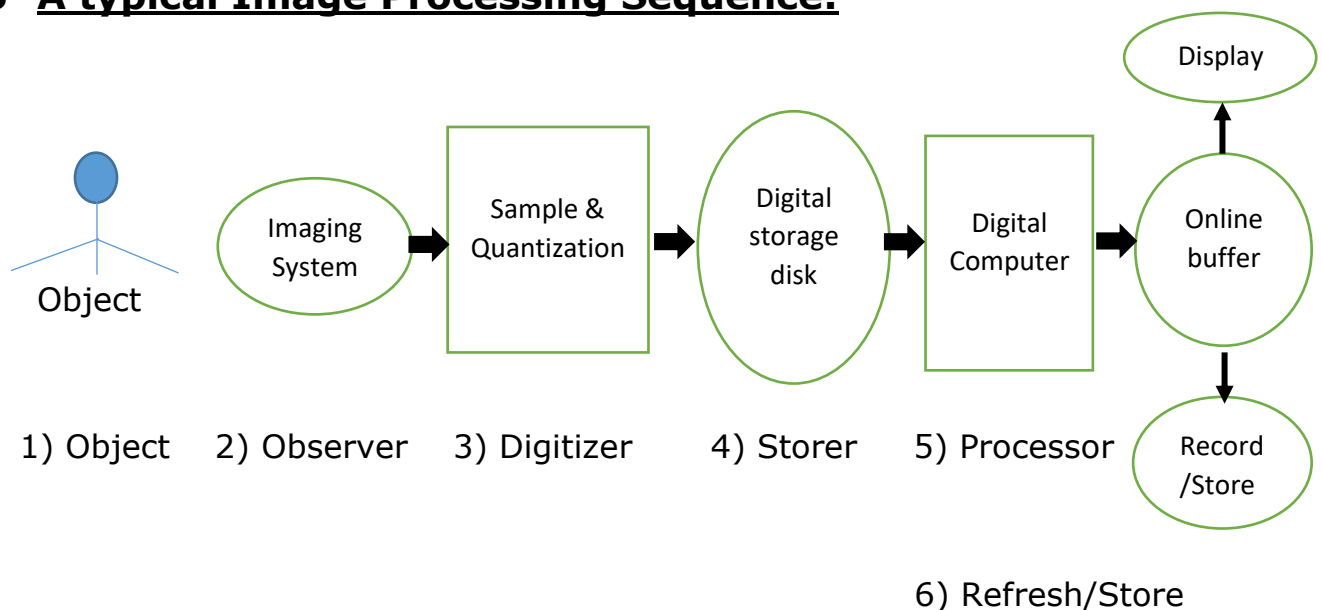
It always digitize the required image by the helps of image preprocessors.

v. Mass Storage:

It is used to store digital image line by line .The storage devices may be magnetic tapes, hard, disk, etc.

vi. Operator Console:

It control the full mechanism of image processing system.

1.6 A typical Image Processing Sequence:**(Fig: Image Processing Sequence)**

In typical image processing sequence involves various components. The image of an object travel through various blocks.

➤ **Object:**

Firstly, the object is placed and sensed by the system and image is formed.

➤ **Imaging System:**

This system sensed the object and creates the image and passed into the sampling and Quantization.

➤ **Digitizer:**

Now the image is created and send over the sampling and quantization for digital image. The image is digitized here with sampling and quantization and passed over Storer.

➤ **Storer:**

The digitized image is passed here to store for further process.

➤ **Processor:**

The digital computer is connected to digital storage disk where all the manipulation of digital image is done.

➤ **Online Buffer:**

Here also the online buffer is also available where the information is stored as a memory.

Here two parts are there one is display and other is store or record.

The image is displayed through the display block which may be monitor, TV, etc.

Finally, the image is recorded in a recording block.

But there may be problem when image is sequentially processed into digital computer.

1.7 **Sampling and Quantization:**

This includes the first elements as a digitization.

➤ **Digitization:**

A physical image is changed into digital image by performing sampling and quantization. This process is also known as digitization.

➤ **Sampling:**

- It is the process where measurements are taken at regular space intervals.
- Digitizing the spatial co-ordinates values (x, y) is called sampling.
- It is the process by which image is formed over a patch (domain) in a continuous domain is managed into a discrete point with integer coordinates.
- Hence sampling can be defined as a selection of discrete location in the continuous 2D space.

➤ **Quantization:**

- It is the process of mapping the measured intensity to one of the finite numbers of discrete levels.
- Hence, digitizing the amplitude values is called quantization.

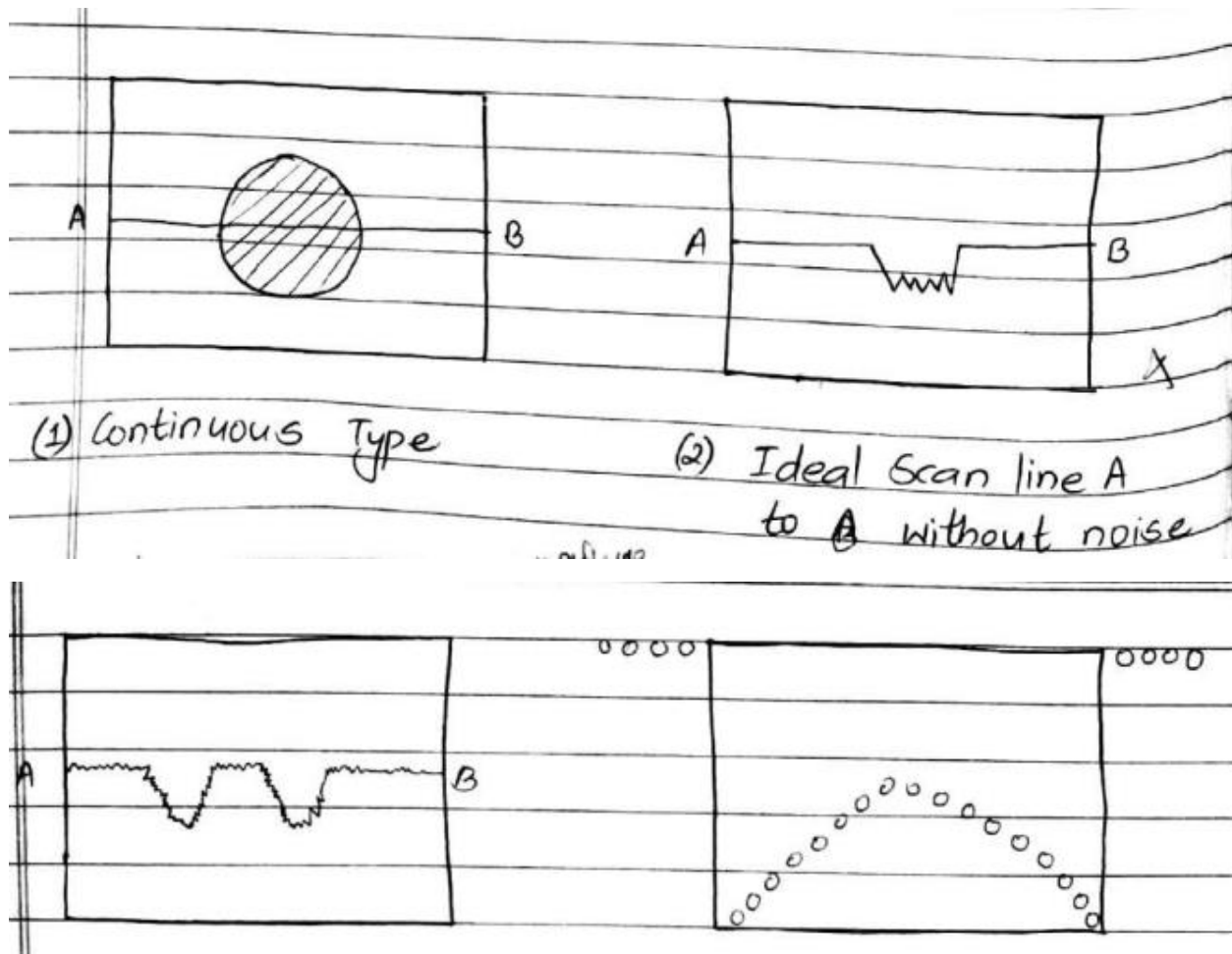
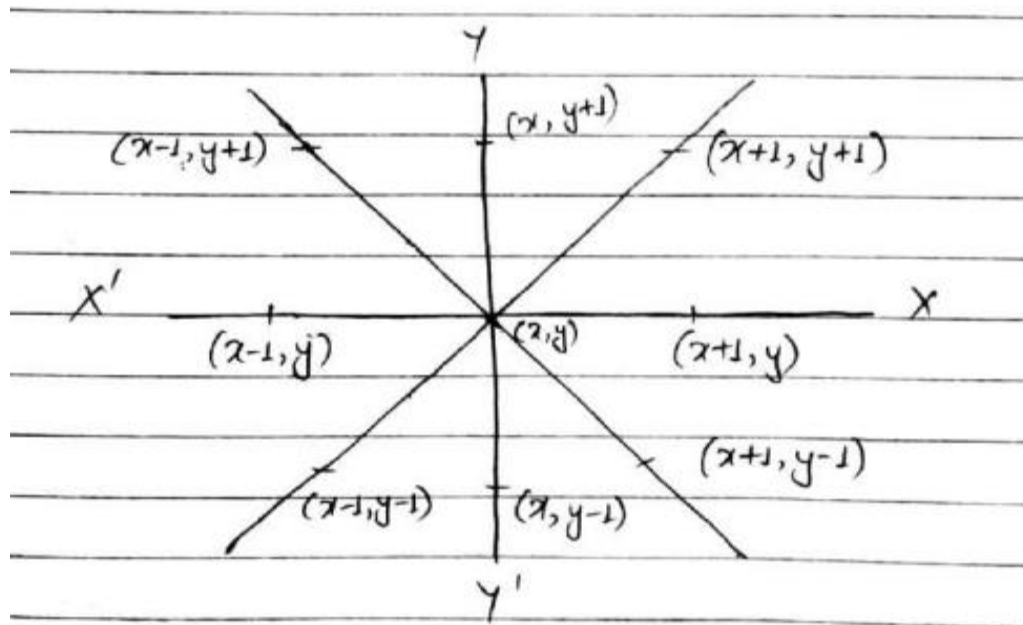


Fig: Sampling and Quantization

1.8. Some Basic relationship like Neighbor's, Connectivity, Distance Measures between Pixels:



(fig: Neighbours of pixel)

A pixel at co-ordinate (x, y) has a four horizontal and four vertical neighbours whose co-ordinates are given by $(x-1, y+1)$, $(x, y+1)$, $(x+1, y+1)$, $(x+1, y)$, $(x+1, y-1)$, $(x, y-1)$, $(x-1, y-1)$, $(x-1, y)$.

These set of pixel is called four neighbours of horizontal and four neighbours of vertical.

Each pixel is at unit distance from (x, y) and called as $N_8(P)$.

Distance Measures:

For pixels p , q , and z , with coordinates (x, y) , (s, t) , and (v, w) , respectively,

D is a distance function or metric if

- (a) $D(p, q) \geq 0$ ($D(p, q) = 0$ iff $p = q$),
- (b) $D(p, q) = D(q, p)$, and
- (c) $D(p, z) \leq D(p, q) + D(q, z)$.

The **Euclidean distance** between p and q is defined as

$$D_e(p, q) = [(x - s)^2 + (y - t)^2]^{\frac{1}{2}}.$$

The **D4 distance (also called city-block distance)** between p and q is defined as

$$D_4(p, q) = |x - s| + |y - t|.$$

In this case, the pixels having a D4 distance from (x, y) less than or equal to some value r form a diamond centered at (x, y) . For example, the pixels with D4 distance. 2 from (x, y) (the center point) form the following contours of constant distance:

$$\begin{array}{ccccc} & & 2 & & \\ & 2 & 1 & 2 & \\ 2 & 1 & 0 & 1 & 2 \\ & 2 & 1 & 2 & \\ & & 2 & & \end{array}$$

The pixels with $D_4 = 1$ are the 4-neighbors of (x, y) .

The **D8 distance (also called chessboard distance)** between p and q is defined as

$$D_8(p, q) = \max(|x - s|, |y - t|).$$

In this case, the pixels with D8 distance from (x, y) less than or equal to some value r form a square centered at (x, y). For example, the pixels with $D_8 \text{ distance} \leq 2$ from (x, y) (the CenterPoint) form the following contours of constant distance:

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

The pixels with $D_8=1$ are the 8-neighbors of (x, y).

Connectivity between pixels:

- It is an important concept in digital image processing.
- It is used for establishing boundaries of objects and components of regions in an image.
- Two pixels are said to be connected:
if they are adjacent in some sense (neighbors' pixels, 4/8/m-adjacency)
if their gray levels satisfy a specified criterion of similarity (equal intensity level)

There are three types of connectivity on the basis of adjacency.

They are:

- 4-connectivity:** Two or more pixels are said to be 4-connected if they are 4-adjacent with each other's.
- 8-connectivity:** Two or more pixels are said to be 8-connected if they are 8-adjacent with each other's.
- m-connectivity:** Two or more pixels are said to be m-connected if they are m-adjacent with each other's.


```

0  1  1
0  1  0
0  0  1

```

Fig: An arrangement of pixels

```

0  1—1
   |
0  1  0
   |
0  0  1

```

Fig: 4-connectivity of pixels

```

0  1—1
   | \
0  1  0
   | /
0  0  1

```

Fig: 8-connectivity of pixels

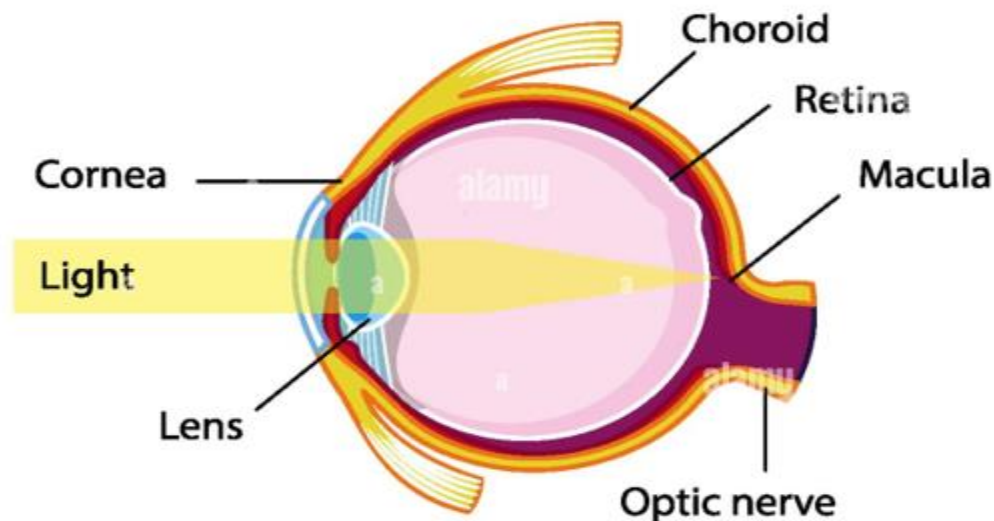
```

0  1—1
   | \
0  1  0
   | /
0  0  1

```

Fig: m-connectivity of pixels

1.9. Elements of Visual Perception :



- **Cornea:** The front transparent part of the sclera is called the cornea. Light enters the eye through the cornea.
- **Iris:** A dark muscular tissue and ring-like structure behind the cornea is known as the iris. The colour of the iris actually indicates the colour of the eye. The iris also helps regulate or adjust exposure by adjusting the iris.
- **Pupil:** A small opening in the iris is known as a pupil. Its size is controlled with the help of iris. It controls the amount of light that enters the eye.
- **Lens:** Behind the pupil, there is a transparent structure called a lens. By the action of ciliary muscles, it changes its shape to focus light on the retina. It becomes thinner to focus on distant objects and becomes thicker to focus on the nearby objects.

- **Retina:** It is a light-sensitive layer that consists of numerous nerve cells. It converts images formed by the lens into electrical impulses. These electrical impulses are then transmitted to the brain through optic nerves.
 - **Optic nerves:** Optic nerves are of two types. These include cones and rods.
1. **Cones:** Cones are the nerve cells that are more sensitive to bright light. They help in detailed central and colour vision.
 2. **Rods:** Rods are the optic nerve cells that are more sensitive to dim lights. They help in peripheral vision.

The human eye nearly a sphere with an average diameter of 14mm to 17mm. The lens of the eye is flexible than the ordinary optical lens.

Let, the observer is looking at a tree 15m height at the distance of 100m. Find the height of retriaval image.

